

Pratik Bairoliya

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EXPERIENCE

- Capital One** Jul. 2026 – Present
Software Engineer II McLean, VA
 - Own the nightly **end-of-day close** — posting, settlement, ledger cutoff across Card and Bank, run on **Temporal**.
 - Lead the **REST API** replacing the vendor's close operator tooling, bringing nightly controls in-house.
 - Monitoring and distributed tracing in **OpenTelemetry** and **Splunk**: alerts page on-call mid-run, not at cutoff.
 - Ship end-of-day services on **Kubernetes** through **Argo CD** and **Helm**, rollback one revision away.
- Capital One** Aug. 2025 – Jul. 2026
Software Engineer I McLean, VA
 - Wrote the ledger's **batch write path** in **Go**, empty repo to production; every nightly cycle runs through it.
 - Designed a **block-allocation ID service**: one call reserves a range, taking per-ID contention off bulk opens.
 - Built the nightly **PySpark/Databricks** reconciliation: day-over-day aggregate diffs alert before a bad cycle ships.
 - Cut data egress on the nightly **99M+ Card account** publish by performance-tuning **Hive** sort on **AWS EMR**.
 - Rolled **proof-of-possession (PoP) tokens** across Card and Bank endpoints, binding each call to a key a thief lacks.
- Capital One** May 2024 – Aug. 2024
Software Engineering Intern Richmond, VA
 - Moved the military rewards catalog off SharePoint into a **Vue.js/Spring Boot/DynamoDB** app: **2.8x** faster.
 - Shipped it to production myself on **Terraform**, **Docker** and **Jenkins** pipelines.
- Fidelity Investments** Jun. 2023 – May 2024
Software Engineer Durham, NC
 - Migrated Fidelity's mainframe files and job calls to **AWS** with **Terraform**, serving **400+** associates.
 - Built the single source of truth for every **mainframe API** as a **Neo4j** graph, cutting **5,000+** duplicate nodes.
 - Delivered the full stack on a 4-person team: **Angular** UI, **Java** services, **S3** metadata sync.

PROJECTS

- Lookback** — automated daily briefing | *Python, Ollama (local LLM), Google APIs, SQLite, launchd* 2026
 - Scheduled job merging Calendar, Gmail, and iMessage into one morning note; runs **100% on-device**.
 - Safe to re-run**: a missed night rebuilds the same note instead of duplicating it.
- Pantry** — receipt-to-groceries tracker | *Python, Apple Vision OCR, Ollama, file watchers* 2026
 - OCR** ingest pipeline turning receipt photos into a pantry list with spend rolled up by category.
 - Idempotent ingest**: rescanning a receipt cannot double-count, so spend totals stay correct.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript/JavaScript, SQL, Cypher, Bash, Git
Cloud & Infrastructure: AWS (EMR, S3, DynamoDB), Linux, Kubernetes, containerization (Docker), workflow orchestration (Temporal), Terraform (IaC), Argo CD, Helm, GitOps, Jenkins, CI/CD
Backend & Data: REST API design, microservices, distributed batch processing, ETL, Apache Spark (PySpark), Hive, Databricks, idempotency, retries, authentication (PoP tokens), Spring Boot, Neo4j
Also: Vue.js, Angular, TypeScript, HTML/CSS, local LLM inference (Ollama), OCR pipelines
Practices: Distributed systems design, observability (Splunk, OpenTelemetry), on-call incident response
Certification: AWS Certified Solutions Architect – Associate (Mar. 2026)

EDUCATION

- North Carolina State University** Aug. 2021 – May 2025
B.S. Computer Science and B.S. Applied Mathematics — GPA 4.0 Raleigh, NC
 - Park Scholar** — full merit scholarship to the top **1%** of applicants for leadership and service.
 - Teaching Assistant** — CSC 221 (Python) and MA 405 (Linear Algebra): labs, office hours, grading.
 - Undergraduate Researcher**, 2022–2024 — built **Python** tooling that automated a faculty group's data analysis.
 - Co-founder, InspireNC** — FIRST Robotics 6908 for **100+** students; taught **Java** to cohorts of 35.

INTERESTS

Competitive Bhangra dance, Taekwondo, weightlifting, and running the Krispy Kreme Challenge.